Chapter 10

BJT DC Biasing Techniques

10.1 Introduction

In the previous lecture, we learned that for a transistor to amplify, it must operate in the **Active Region**. But how do we ensure it stays there?

Biasing is the process of applying external DC voltages to set the transistor's operating point—specifically the values of Collector Current (I_C) and Collector-Emitter Voltage (V_{CE}). This fixed DC point is called the **Quiescent Point (Q-Point)**.

A good biasing circuit must provide:

1. A Q-point in the center of the active region (to allow maximum signal swing without clipping).
2. **Stability:** The Q-point should not shift significantly if the temperature changes or if the transistor is replaced with another having a different β.

Learning Objectives

By the end of this chapter, you should be able to:

- Explain the concept of the Q-Point and the DC Load Line.
- Analyze Fixed Bias, Emitter Bias, and Voltage Divider Bias circuits.
- Calculate bias currents (I_B, I_C) and voltages (V_{CE}, V_E) for any configuration.
- Evaluate the stability of different biasing techniques against β variations.

10.2 The DC Load Line

The DC Load Line is a graphical method to visualize the possible operating points of a transistor circuit. It is derived from the output loop equation (Kirchhoff's Voltage Law).

Consider a simple Common Emitter circuit with a collector resistor R_C and supply V_{CC}. Applying KVL to the output loop:

$$V_{CC} = I_C R_C + V_{CE} \tag{10.1}$$

Rearranging for I_C (equation of a straight line $y = mx + c$):

$$I_C = -\frac{1}{R_C} V_{CE} + \frac{V_{CC}}{R_C} \tag{10.2}$$

10.2.1 Plotting the Load Line

1. **Y-Intercept (Saturation Point):** Set $V_{CE} = 0$.

$$I_{C(sat)} = \frac{V_{CC}}{R_C}$$

   This represents the maximum possible current (Transistor acting as a short circuit).

2. **X-Intercept (Cutoff Point):** Set $I_C = 0$.

$$V_{CE(cutoff)} = V_{CC}$$

   This represents the maximum voltage (Transistor acting as an open switch).

10.3 Fixed Bias Configuration

This is the simplest biasing circuit. A single resistor (R_B) connects the Base directly to the supply voltage V_{CC}.

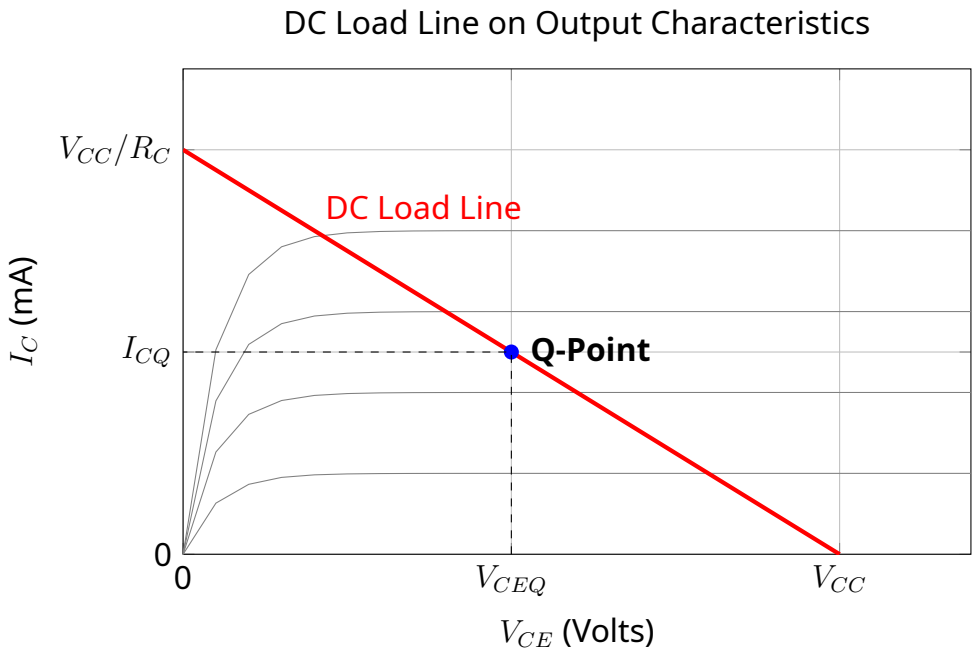


Figure 10.1: The intersection of the Load Line and the transistor curve (I_B) is the Q-Point.

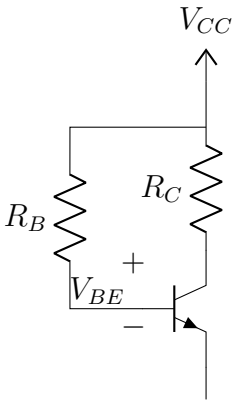


Figure 10.2: Fixed Bias Circuit.

10.3.1 Circuit Analysis

Input Loop (Base-Emitter): Applying KVL:

$$V_{CC} - I_B R_B - V_{BE} = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} \tag{10.3}$$

Since V_{CC} and V_{BE} (0.7V) are constant, I_B is constant (fixed).

Output Loop (Collector-Emitter):

$$I_C = \beta I_B$$

$$V_{CE} = V_{CC} - I_C R_C \tag{10.4}$$

10.3.2 Stability Issue

The collector current is $I_C = \beta I_B$. If the temperature rises, or if we replace the transistor, β can change significantly (e.g., from 100 to 200).

- Since I_B is fixed, if β doubles, I_C doubles.
- The Q-point shifts drastically towards saturation.
- **Conclusion:** Fixed bias has very **poor stability**. It is rarely used in linear amplifiers.

10.4 Emitter-Stabilized Bias

To improve stability, a resistor (R_E) is added to the Emitter terminal.

10.4.1 Circuit Analysis

Input Loop:

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

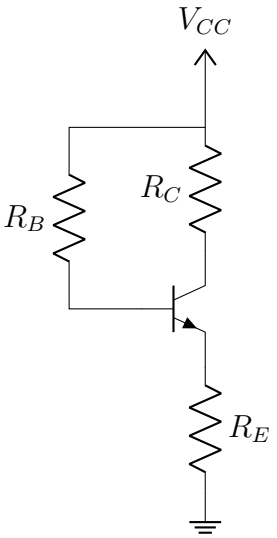


Figure 10.3: Emitter-Stabilized Bias Circuit.

Substitute $I_E = (\beta + 1)I_B$:

$$V_{CC} - V_{BE} = I_B R_B + (\beta + 1) I_B R_E$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E} \tag{10.5}$$

Equation Interpretation: The term $(\beta + 1)R_E$ is the resistance of the emitter reflected back to the base. It is much larger than R_E alone.

10.4.2 Stability Mechanism

If temperature rises $\rightarrow$ β increases $\rightarrow$ I_C attempts to increase.

- If I_C increases, I_E increases.
- The voltage drop across the emitter resistor ($V_E = I_E R_E$) increases.
- Since V_B is fixed by R_B, the voltage across the junction ($V_{BE} = V_B - V_E$) **decreases**.
- A lower V_{BE} reduces the Base current I_B.
- A lower I_B pulls I_C back down.

This is **Negative Feedback**. It automatically opposes changes in current, making the circuit more stable.

10.5 Voltage Divider Bias

This is the standard biasing method used in professional circuit design. It uses a voltage divider (R_1 and R_2) to set a fixed voltage at the Base, independent of the transistor's β.

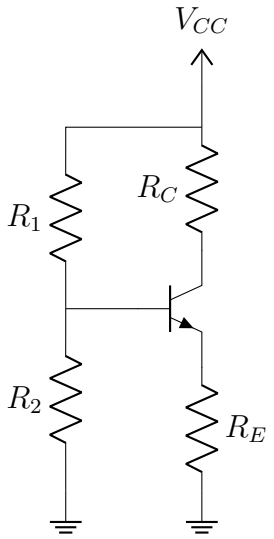


Figure 10.4: Voltage Divider Bias Circuit.

10.5.1 Approximate Analysis

If the input impedance looking into the base (βR_E) is much larger than R_2 (typically $10\times$), we can assume the base current I_B is negligible compared to the current flowing through R_1 and R_2.

Step 1: Calculate Base Voltage (V_B) Using the voltage divider rule:

$$V_B = \frac{R_2}{R_1 + R_2} V_{CC} \tag{10.6}$$

Note: This voltage depends only on external resistors, not on the transistor β. This is why it is stable.

Step 2: Calculate Emitter Voltage (V_E)

$$V_E = V_B - V_{BE} \quad (\text{Usually } V_B - 0.7) \tag{10.7}$$

Step 3: Calculate Emitter Current (I_E)

$$I_E = \frac{V_E}{R_E} \tag{10.8}$$

Step 4: Calculate Collector Parameters

$$I_C \approx I_E$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

Example 1: Voltage Divider Calculation

Given: $V_{CC} = 10$V, $R_1 = 10k\Omega$, $R_2 = 2.2k\Omega$, $R_C = 1k\Omega$, $R_E = 500\Omega$. $\beta = 100$. Find I_C and V_{CE}.

Solution: 1. Find V_B:

$$V_B = \frac{2.2k}{10k + 2.2k} \times 10 = \frac{2.2}{12.2} \times 10 = 1.8 \text{ V}$$

2. Find V_E:

$$V_E = 1.8 - 0.7 = 1.1 \text{ V}$$

3. Find I_E:

$$I_E = \frac{1.1 \text{ V}}{500\Omega} = 2.2 \text{ mA}$$

So, $I_C \approx 2.2$ mA. 4. Find V_{CE}:

$$V_{CE} = 10 - 2.2m(1k + 500) = 10 - 2.2m(1.5k) = 10 - 3.3 = 6.7 \text{ V}$$

The Q-point is (6.7V, 2.2mA).

10.5.2 Exact Analysis (Thevenin's Theorem)

If the assumption $\beta R_E \geq 10R_2$ is *not* met (i.e., R_2 is large or β is small), I_B is not negligible. We must use Thevenin's theorem on the input side.

- **Thevenin Resistance (R_{TH}):** Parallel combination of R_1 and R_2 (short V_{CC} to

ground).

$$R_{TH} = R_1 || R_2 = \frac{R_1 R_2}{R_1 + R_2}$$

- **Thevenin Voltage (V_{TH}):** The voltage divider output.

$$V_{TH} = \frac{R_2}{R_1 + R_2} V_{CC}$$

Now, applying KVL to the simplified base loop:

$$V_{TH} - I_B R_{TH} - V_{BE} - I_E R_E = 0$$

Substitute $I_E = (\beta + 1) I_B$:

$$I_B = \frac{V_{TH} - V_{BE}}{R_{TH} + (\beta + 1) R_E} \tag{10.9}$$

Once I_B is found, we find $I_C = \beta I_B$ and proceed as normal.

10.6 Review Questions

Part A: Conceptual Questions

1. Define "Quiescent Point" (Q-Point). Why is it usually placed in the center of the DC Load Line?
2. Explain the mechanism of "Thermal Runaway" in a transistor and how Emitter-Stabilized biasing prevents it.
3. Why is Fixed Bias configuration rarely used in linear amplifiers despite its simplicity?
4. In a Voltage Divider bias circuit, what condition must be met to use the "Approximate Analysis" method?
5. Sketch the DC Load Line for a circuit with $V_{CC} = 12$V and $R_C = 2k\Omega$. Label the Saturation and Cutoff points.
6. What is the effect of decreasing R_B in a Fixed Bias circuit on the Q-point?
7. Why is the capacitor used in an amplifier circuit treated as an open circuit during DC analysis?
8. What is the meaning of "Beta Independent" biasing?
9. Explain how adding an Emitter Resistor (R_E) reduces the voltage gain of an amplifier while increasing stability.
10. What happens to the Q-point of an NPN transistor if the base-emitter junction becomes open?

Part B: Advanced Calculations & Analysis

11. **Fixed Bias Analysis:** A fixed bias circuit has $V_{CC} = 12$V, $R_B = 240k\Omega$, and $R_C = 2.2k\Omega$. The transistor $\beta = 50$.
    (a) Calculate I_B, I_C, and V_{CE}.

(b) If the transistor is replaced by a new one with $\beta = 100$, calculate the new Q-point (I_C, V_{CE}).

(c) Comment on the shift in the Q-point.

12. **Emitter Bias Design:** Design an emitter-stabilized bias circuit to operate at $I_C = 2$ mA and $V_{CE} = 10$V. The supply voltage is $V_{CC} = 20$V. Assume $\beta = 100$. *(Hint: Rule of thumb, let $V_E \approx 0.1V_{CC}$. Use standard resistor equations).*

13. **Voltage Divider (Exact vs Approx):** A voltage divider circuit has $V_{CC} = 18$V, $R_1 = 33k\Omega$, $R_2 = 8.2k\Omega$, $R_E = 1k\Omega$, and $R_C = 1.5k\Omega$.

(a) Calculate I_C and V_{CE} using the **Approximate** method.

(b) Calculate I_C and V_{CE} using the **Exact (Thevenin)** method assuming $\beta = 50$.

(c) Calculate the percentage error between the two methods. Is the approximation valid here?

14. **Saturation Analysis:** Using the circuit in Question 13 (with $\beta = 50$), determine the minimum value of R_C that would push the transistor into the Saturation region.

15. **Load Line Plotting:** For the circuit in Question 13:

(a) Determine $I_{C(sat)}$ and $V_{CE(cutoff)}$.

(b) Write the equation of the DC Load Line.

(c) Plot the Load Line and mark the Q-point calculated in 13(b).